

Normal Form Games

Formal model NF game: tuple (N, A, u) : $N = \{1..n\}$ agents; $A = A_1 \times \dots \times A_n$ action profiles; $u = (u_1, \dots, u_n)$, $u_i : A \rightarrow \mathbb{R}$. A profile $a = (a_i, a_{-i})$; $a_{-i} = \text{others' actions}$. Payoff $u_i(a_i, a_{-i})$. Finite + 2 players $\Rightarrow$ payoff **matrix** (row=P1, col=P2, cell = (u_1, u_2)).

Modeling a scenario Checklist: + How many agents N ? + Each action set A_i ? + Payoffs u_i (one number per agent per profile)? Discrete $\rightarrow$ matrix; continuous $A_i \rightarrow$ calculus.

Best response & dominance **Best response:** $a_i \in \text{BR}_i(a_{-i})$ iff $u_i(a_i, a_{-i}) \geq u_i(a'_i, a_{-i}) \forall a'_i$. **Strictly dominated:** a_i str. dom. by a'_i if $u_i(a'_i, a_{-i}) > u_i(a_i, a_{-i})$ for every a_{-i} . **Weakly dominated:** $\geq$ for all a_{-i} and $>$ for at least one.

IESDS — procedure

- 1. Find any **strictly** dominated action for some agent.
- 2. Delete it; reduced game.
- 3. Repeat (new dominations may appear after deletions).
- 4. Stop when none remain $\rightarrow$ surviving profile(s) = prediction.

Use **STRICT** dominance only. Eliminating **weakly** dominated actions can destroy real NE and is order-dependent.

Pure-strategy Nash equilibrium **Pure NE:** profile a^* s.t. no agent gains by unilateral deviation: $u_i(a_i^*, a_{-i}^*) \geq u_i(a_i, a_{-i}^*) \forall i, \forall a_i$. **Underline / best-response method:** + per column, mark P1's best payoff(s); + per row, mark P2's best payoff(s); + cell with **both** marked = pure NE (may be 0, 1, or many).

Mixed strategies & mixed NE **Mixed strategy:** prob. distribution σ_i over A_i ; pure = degenerate. 2-action: $(q, 1-q)$. Expected payoff $u_i(\sigma) = \sum_a (\prod_j \sigma_j(a_j)) u_i(a)$. **Mixed NE:** each σ_i^* maximizes expected payoff vs σ_{-i}^* . **Nash's theorem:** every finite game has ≥ 1 NE (possibly mixed).

Indifference principle — find mixed NE (2x2)

- 1. Let opponent play action w.p. p . + Write your EU(act₁), EU(act₂) as fns of p . + Set them **equal** (indifference) $\rightarrow$ solve p . + Repeat for other player.

The p that solves "make ME indifferent" is the prob. the **OPPONENT** must play. Mixing probs that make the opponent indifferent are what **YOU** play. Easy to swap players.

Canonical 2x2 games

Prisoner's Dilemma (advertising): C str. dominates $\rightarrow$ unique (Adv, Adv) = (3, 3), both worse than (5, 5).

	Adv	Don't
Adv	3, 3	6, 2
Don't	2, 6	5, 5

Matching Pennies (zero-sum, outguess): **no pure NE;** mixed NE $(\frac{1}{2}, \frac{1}{2}), (\frac{1}{2}, \frac{1}{2})$.

	H	T
H	-1, 1	1, -1
T	1, -1	-1, 1

Battle of Sexes (coordination): 2 pure NE (Ballet,Ballet), (Fight,Fight); mixed: man = $(\frac{1}{3}, \frac{2}{3})$, woman = $(\frac{2}{3}, \frac{1}{3})$ over (Ballet,Fight) — players mix with **different** probs.

	Ballet	Fight
Ballet	1, 2	0, 0
Fight	0, 0	2, 1

BoS mixed: woman plays Ballet w.p. p ; man indiff $1 \cdot p = 2(1-p) \Rightarrow p = \frac{2}{3}$.

Cournot duopoly (continuous) 2 firms, homogeneous good, no collusion, choose quantities $q_i \in [0, \infty)$ simultaneously (one-shot). Inverse demand $P(Q) = a - Q$ for $Q < a$ ($Q = q_1 + q_2$), const. marg. cost $c < a$, no fixed cost. Profit: $\pi_i(q_i, q_j) = q_i[a - (q_i + q_j) - c]$. **FOC** (firm 1): $\frac{\partial \pi_1}{\partial q_1} = a - 2q_1 - q_2 - c = 0$. **Best-response (reaction) fns:** $b_1(q_j) = \frac{1}{2}(a - q_j - c)$. NE = intersection of b_1, b_2 :

$q_1^* = q_2^* = \frac{1}{3}(a - c)$ — symmetric Cournot-Nash equilibrium.

Discrete version: enumerate $Q_i \in \{0..5\}$, $P = \max(0, 12 - 2Q)$, $\pi_i = (P - 1)Q_i$, then apply BR/NE method.

Interpreting mixed strategies Confuse/outguess opponent; uncertainty about other's action; limiting frequency in repeated play; population state (fraction playing each pure strategy). Apps: penalties, patrols, audits — "unpredictable in a precise way."

Extensive Form Games

Tree/sequential representation: makes **temporal structure** explicit (normal-form = simultaneous, one-shot). Treat perfect-info first, then imperfect.

Perfect-info game (tuple) $G = (N, A, H, Z, \chi, \rho, \sigma, u)$: N agents; A actions; H choice nodes; Z terminals; $\chi : H \rightarrow 2^A$ actions-at-node; $\rho : H \rightarrow N$ who-moves; $\sigma : H \times A \rightarrow H \cup Z$ successor (each node unique predecessor $\Rightarrow$ tree); $u_i : Z \rightarrow \mathbb{R}$.

- node = choice, edge = action, leaf = payoff vector.

Pure strategy = complete contingent plan Action at **every** own node (perfect) / **every own info set** (imperfect) — even unreachable ones. Pure strategies of i :

$$\prod_{h \in H, \rho(h)=i} \chi(h) \quad (\text{imperfect: } \prod_{I_{i,j} \in I_i} \chi(I_{i,j}))$$

Strategy specifies actions even at nodes never reached. This causes normal-form **redundancy** and lets some NE rest on **non-credible threats**.

Extensive $\rightarrow$ Normal form (build matrix)

- 1. Rows = S_1 , cols = S_2 (full strategy sets, incl. contingent actions).
- 2. For each pair (s_1, s_2) : follow the path it induces from root to a leaf.
- 3. Cell = payoff vector at that leaf.
- 4. Find NE the usual (normal-form) way.

Running game: 1 at root $A|B$; 2 plays $C|D$ after A , $E|F$ after B ; 1 plays $G|H$ after F . Leaves: $AC(3,8)$, $AD(8,3)$, $BE(5,5)$, $BFG(2,10)$, $BFH(1,0)$, $DE(4,9)$, $AEH(3,1)$, $AG(1,0)$, $DF(1,0)$, $DF(1,0)$.

1\2	CE	CF	DE	DF
(A,G)	3,8	3,8	8,3	8,3
(A,H)	3,8	3,8	8,3	8,3
(B,G)	5,5	2,10	5,5	2,10
(B,H)	5,5	1,0	5,5	1,0

Redundancy: 16 normal-form cells vs only 5 leaves — $(A,G) = (A,H)$ ($G|H$ no node unreached after A). Pure NE: $\{(AG), (CF)\}, \{(AH), (CF)\}, \{(BH), (CE)\}$. **Thm:** every finite perfect-info game has a **pure-strategy NE** (turns + full observation $\Rightarrow$ no randomness needed).

Subgame-perfect equilibrium (SPE) **Subgame at h :** restriction of G to descendants of h (G itself is a subgame). **SPE:** profile s s.t. its restriction to **every** subgame is a NE of that subgame. Rules out non-credible threats.

- Every SPE is a NE; **not** every NE is SPE; ≥ 1 SPE always exists.
- To **disprove** SPE: find one subgame (usually bottom-most) where restricted profile is not a NE. Above: $\{(BH), (CE)\}$ fails (1 plays H , but $G > 1 > 0$) $\Rightarrow$ only $\{(AG), (CF)\}$ is SPE.

Backward induction (computes a SPE) Single depth-first pass, linear in tree size.

- 1. Recurse to terminal; return its payoff vector.
- 2. At node h , evaluate each child via recursion.
- 3. Pick child maximizing the **mover $\rho(h)$'s coordinate**; return that whole vector up.

BI(h): if $h \in Z$ return $u(h)$
best $\leftarrow -\infty$
for $a \in \chi(h)$: $c \leftarrow \text{BI}(\sigma(h,a))$
if $c[p(h)] > \text{best}[p(h)]$: best $\leftarrow c$
return best

Compare on the moving agent's component $\rho(h)$ only; **best** is a full vector $(u_1, \dots, u_n)$.

Zero-sum: BI = **minimax** (one number/node, speed up with **alpha-beta pruning**).

Imperfect information + info sets Add I : $G = (\dots, u, I, I_i)$ partitions i 's nodes into **information sets**. Nodes in same set are **indistinguishable**; must have **same action set** $\chi(I_{i,j})$ (else distinguishable).

- Models **simultaneous** moves / hidden info (PD, Poker) — perfect-info trees **cannot**.

Converting a normal-form (simultaneous) game needs an info set linking the 2nd mover's nodes — else the tree wrongly lets him observe move 1.

- Normal $\leftrightarrow$ imperfect-info mappings both exist but are **not bijective**; round-trip gives a bigger but **strategically equivalent** game (same strategy space + equilibria).

Perfect vs imperfect & randomization

Perfect info	Imperfect info
knows current node + all past moves	some own-nodes indistinguishable
pure-strat NE exists	may need mixed (coordination)
BI direct	only over proper subgames

- **Mixed:** one lottery over pure strategies at start. **Behavioral:** independent draw at each info set.

- **Kuhn (1953):** under **perfect recall** mixed $\leftrightarrow$ behavioral are equivalent. Under **imperfect recall** they can differ (behavioral may do strictly better).

Bayesian Games

Bayesian game: game of **incomplete information**: agents are uncertain about **which game is played**. Uncertainty = prob. distribution over possible games. Two assumptions: (1) games differ only in **payoffs** (same N , same action sets); (2) beliefs are **posteriors** from conditioning a **common prior** on private signals.

Two equivalent definitions

- **Info-sets:** (N, G, P, I) — agents, set of games (same actions, diff payoffs), common prior $P \in \Pi(G)$, partitions $I = (I_1, \dots, I_N)$ of G (each $I_i = i$'s indistinguishable-game classes).
- **Types:** (N, A, Θ, p, u) — $A = A_1 \times \dots \times A_n$; type space $\Theta = \Theta_1 \times \dots \times \Theta_n$; common prior $p : \Theta \rightarrow [0, 1]$; $u_i : A \times \Theta \rightarrow \mathbb{R}$.

- **Equivalence:** each partition cell $\leftrightarrow$ one type $(I_{i,k} \rightarrow \theta_{i,k})$; prior over games $P \rightarrow$ prior over type profiles p .

Type θ_i : agent i 's **private info** (indexes its payoff fn), **i knows own type**, not others'. (θ, a) fully determines all payoffs.

Strategies & timing

- **Pure strategy:** $\alpha_i : \Theta_i \rightarrow A_i$ — maps every type of i to an action (contingent plan, one action per type). Equilibrium checked **per type**.
- **Timing** of **info:** **ex-ante** knows no type (not even own); **interim** knows own θ_i only; **ex-post** knows all types.

Interim expected utility

$$EU_i(\alpha \text{ mid } \theta_i) = \sum_{\theta_{-i} \in \Theta_{-i}} p(\theta_{-i} \text{ mid } \theta_i) u_i(\alpha_i(\theta_i, \theta_{-i}))$$

$p(\theta_{-i} \text{ mid } \theta_i) =$ **posterior** over others' types given own type; α evaluated = action each type plays.

Bayesian Nash Equilibrium (BNE) Profile α s.t. for each i and each $\theta_i \in \Theta_i$:

$$\alpha_i \in \arg \max_{\alpha'_i} EU_i(\alpha'_i, \alpha_{-i} \text{ mid } \theta_i)$$

Every **type** best-responds, expecting over **both actions and types** of others.

BNE is an **interim** concept (each type best-responds). Use posterior $p(\theta_{-i} \text{ mid } \theta_i)$, NOT raw joint $p(\theta_i, \theta_{-i})$ — only equal up to normalization when types independent. A type with a **strictly dominant action** must play it in any BNE.

Compute BNE — procedure

- 1. List each agent's types; pure strat = action choice per type.
- 2. Per type, check **strict dominance**: if one action beats all others for that type regardless of others, fix it.
- 3. For remaining types, compute $EU_{i_i}(\text{mid } \theta_{i_i})$ via posterior over θ_{-i} ; pick best response.
- 4. Profile is BNE iff all type best-responses hold simultaneously.

Canonical ex: **Firm hires Worker** $N = \{F, W\}$; $A_F = \{\text{hire, dont}\}$, $A_W = \{\text{work, shirk}\}$; $\Theta_F = \{t_f\}$, $\Theta_W = \{\text{high, low}\}$; prior $p(\text{high}) = p$.

$\theta_W = \text{high}$	work	shirk
hire	1, 2	0, 1
dont	0, 0	0, 0

$\theta_W = \text{low}$	work	shirk
hire	1, 1	-1, 2
dont	0, 0	0, 0

Worker high: $\text{work}(2) > \text{shirk}(1)$. Worker low: $\text{shirk}(2) > \text{work}(1)$. So $\alpha_W^* = (\text{work, shirk})$. Firm vs α_F^* : $EU_F(\text{hire}) = p(1) + (1-p)(-1) = 2p - 1$, $EU_F(\text{dont}) = 0$. Hire iff $p \geq \frac{1}{2}$. At $p = \frac{3}{4}$, $\frac{1}{2} > 0 \Rightarrow$ hire $\Rightarrow \alpha^*$ is BNE.

Modeling tip Nature draws types from prior; each agent observes own signal, then acts. Hide Nature's choice from uninformed agents via an **information set** (extensive form) $\leftrightarrow$ private types.

Repeated Games

A **stage game** = a normal-form game played repeatedly. Key questions: can agents **observe** others' actions, **remember** the past, and what is **total utility**?

Finitely-Repeated $G(T)$

- Modelled as **extensive-form, imperfect info** (don't see this round's simultaneous move, but recall past). Total payoff = **sum** of stage payoffs.

- **Proposition:** If stage game G has a **unique** NE, then for any finite T , $G(T)$ has a **unique** outcome: that NE played **every** stage (by backward induction).

- **Backward induction:** last stage has no future $\Rightarrow$ play stage-NE; knowing that, 2nd-last unravels too, back to round 1.

- **Finitely-repeated PD:** cooperation is **impossible**. Rational agents defect every round. "Always defect" is a NE.

Infinitely-Repeated Infinite tree; naive sum = ∞ , so redefine utility over payoffs $r_1, r_2, \dots$:

$$\text{Average: } \lim_{k \rightarrow \infty} \sum_{j=1}^k \frac{r_j}{k} \quad \text{Discounted: } \sum_{j=1}^{\infty} \beta^j r_j, \quad 0 < \beta < 1$$

Geometric series (the workhorse for infinite streams): $\sum_{t=0}^{\infty} \beta^t = \frac{1}{1-\beta}$. So a constant c every period from **now** = $\frac{c}{1-\beta}$; starting **next** period = $\beta \frac{c}{1-\beta}$.

- **Discount factor β :** weight on the future. Low β = impatient (cares about near term); high β = patient. Also = prob. game continues.
- Pure strategy maps history to action: $s_i : H \rightarrow A_i$, $H = U_i H^i$.
- **Tit-for-tat:** cooperate; if opp defects, defect one round then forgive.
- **Grim trigger:** cooperate; if opp ever defects, defect forever.
- **SPE:** NE in every subgame (after every history). Repeating a stage-game NE is **always** an SPE.

Folk Theorem Let $a =$ NE of stage game G . If a' gives $u_i(a') > u_i(a)$ for all i (Pareto-bets the NE), then $\exists \beta \in (0, 1)$ s.t. for patient enough players ($\beta_i \geq \beta$), there is an **SPE** of the

infinite game playing a' every period. Enforced by **trigger** strategies.

Sustaining cooperation: deviation algebra Trigger: play a' ; if anyone deviates, play NE a forever (punishment is SPE since a is stage-NE). Define one-shot temptation and per-period punishment loss:

$$M = \max_{i, a'_i} u_i(a'_i, a_{-i}^*) - u_i(a^*), \quad m = \min_{i, a'_i} u_i(a^*) - u_i(a')$$

Net gain from deviating = $M - m \frac{\beta}{1-\beta_i}$. Deviation **not** worth it iff:

$$M - m \frac{\beta_i}{1-\beta_i} \leq 0 \Leftrightarrow \beta_i \geq \frac{M}{M+m} \quad \forall i$$

Canonical example — PD, sustain (C, C)

	C	D
C	3, 3	0, 5
D	5, 0	1, 1

Want $(C, C) = 3$; stage NE = $(D, D) = 1$. Compare discounted streams (first term **undiscounted**):

- Cooperate: $\frac{3}{1-\beta}$ Defect (then punished at 1): $5 + \frac{\beta}{1-\beta} - 2 \geq 0 \Rightarrow \beta \geq \frac{1}{2}$.
- **Interpretation:** must value tomorrow at least half as much as today.
- Bigger temptation (D payoff 10 not 5): diff = $-7 + \beta \frac{2}{1-\beta} \Rightarrow \beta \geq \frac{7}{9}$ (more patience needed).

Trap: finite + unique NE $\Rightarrow$ no cooperation (unravels from last round); infinite + high $\beta \Rightarrow$ cooperation possible. TFT forgives after 1 round; Trigger punishes forever (Folk proof uses Trigger).

Multiagent Learning

Stochastic (Markov) Game Tuple (Q, N, A, P, r) : Q states/games, $N = n$ agents, $A = A_1 \times \dots \times A_n$ joint actions, $P(q, a, \hat{q}) \in [0, 1]$ transition, $r_i : Q \times A \rightarrow \mathbb{R}$. Generalizes: 1 state $\Rightarrow$ repeated game; 1 agent $\Rightarrow$ MDP. **Learning taxonomy** (who do you model?)

Type	Q-function / how others treated
Independent	$Q_i(s_i, a_i)$; others = environment
Centralized	one entity decides all
JAL	$Q_i(s, a_1..a_n)$; need solution concept C
JAL-AM	JAL + opponent model $\hat{\pi}_j$

Decision tree: intentions? No $\rightarrow$ Indep. Yes $\rightarrow$ single agent? Yes $\rightarrow$ Central. No $\rightarrow$ model others? Yes $\rightarrow$ JAL-AM. No $\rightarrow$ JAL.

Q size: Indep N_a (own action), JAL $N_a \times N_a$ ($|S| \times |A|^n$, exp in n ; n tables). **max / ϵ -greedy only valid for Independent** — JAL best value depends on others $\Rightarrow$ use C .

Q-learning update (Independent)

$$Q(s_i, a_i) \leftarrow Q(s_i, a_i) + \alpha \left[r + \gamma \max_{b_i} Q(s_i, b_i) - Q(s_i, a_i) \right]$$

Flaw: others learning $\Rightarrow$ transition model **non-stationary**, breaks convergence. Claus&Boutilier: may reach NE but **not Pareto-optimal**.

JAL value iteration / solution concept C $V_i(s) := C(Q_1(s, a_1), \dots, Q_n(s, a_n))$; $Q_i(s, a_i) := R_i(s, a_i) + \gamma \sum_{s'} P(s' | s, a_i) V_i(s')$

minimaxQ: zero-sum only; LP for maximin: $\pi[s, \cdot] = \arg \max_{\sigma'} \min_{\sigma''} \sum_{s'} \pi[s, a'] Q[s, a', \sigma'']$. $V[s] =$ that min. **NashQ:** general-sum; bootstrap with NashQ($s^{(0)}$) = value of stage-game NE at $s^{(0)}$, not max.

Fictitious Play (repeated games) — RECIPE Model-based; needs **only own payoff matrix**. Belief = empirical freq of opponent's actions.

$$P(a) = w(a) / \sum_{a'} w(a')$$

- 1. Init beliefs = action counts w (**must be nonzero**, not all-0; sensitive to prior).
- 2. Best-respond to assessed mixed strategy P .
- 3. Observe opponent's real play, increment its $w(a)$. Repeat.

Worked ex: Opp plays C,C,N,C,N $\Rightarrow w = (3, 2) \Rightarrow (0.6, 0.4)$. Matching Pennies: empirical dist $\rightarrow (0.5, 0.5) =$ unique NE (per-round still oscillates).

JAL-AM (fictitious play for stochastic games)

$$\pi_j(a_j | s) = \frac{C(s, a_j)}{\sum_{a'_j} C(s, a'_j)}; \quad \text{AV}_i(s, a_i) = \sum_{a_{-i}} Q_i(s, (a_i, a_{-i})) \prod_{j \neq i} \pi_j(a_j | s)$$

Act: $\arg \max_{a_i} \text{AV}_i(\epsilon\text{-greedy})$; update Q_i with $\gamma \max_{a_i} \text{AV}_i(s', a_i)$.

Stag Hunt / coordination Game collection in stochastic games (PD, Stag Hunt, Battle of Sexes). Reward depends on others' actions (bears: both right = +5 each, both left = +1, split: mover-left +2, mover-right 0) $\Rightarrow$ motivates observing state (distributed) then actions (JAL).

Auctions

Auction: mechanism allocating scarce resources among self-interested agents. All bidders know **number of bidders**.

Four Canonical Auctions

Type	Format	Winner pays	Bids
English	open ascending	own (highest) bid	many, public
Dutch	open descending clock	price at accept	1 accept, public
1st-price SB	sealed 1-shot	own highest bid	1, private
2nd-price SB (Vickrey)	sealed 1-shot	2nd-highest bid	1, private

- English:** auctioneer opens at reserve; bidders should ascend; highest standing bid wins if unchallenged. Art, wine.
- Dutch:** clock starts high, drops each step until someone accepts and pays that ask. Flowers, produce.
- Strategic links: **English $\approx$ 2nd-price** (truthful), **Dutch $\approx$ 1st-price** (shade).

First-Price Sealed-Bid Highest bidder wins, pays own bid $\rightarrow$ incentive to **shade** ($b_i < v_i$). **No dominant strategy** (best response depends on others). Tradeoff: higher b = win more often but less profit $v_i - b_i$.

BNE of 1st-Price (risk-neutral, v_i iid $\sim U(0,1)$)

General N bidders:
$b_i = \frac{N-1}{N} v_i$
As $N \rightarrow \infty$, $\frac{N-1}{N} \rightarrow 1$ (shade less w/ more rivals). $N = 2$: $b_i = \frac{1}{2} v_i$.

Derivation ($N = 2$, given $b_2 = \frac{1}{2} v_2$): bidder 1 wins iff $b_2 < b_1 \Leftrightarrow b_2 < 2b_1$. $\mathbb{E}[u_1] = P(v_2 < 2b_1)(v_1 - b_1) = F(2b_1)(v_1 - b_1)$ $U(0,1) \Rightarrow F(x) = x$, so $\mathbb{E}[u_1] = 2b_1(v_1 - b_1) = 2v_1 b_1 - 2b_1^2$. FOC: $\frac{\partial}{\partial b_1} = 2v_1 - 4b_1 = 0 \Rightarrow b_1 = \frac{v_1}{2}$.

Second-Price (Vickrey) Highest wins, pays **2nd-highest** bid. **Truthful $b_i = v_i$ is weakly dominant** (optimal regardless of others) and a NE.

Trap: 2nd-price winner pays the second-highest bid, NOT own bid. Ex: bids 100, 95, 80 $\rightarrow$ winner pays 95 , not 100 . 1st-price winner of same bids pays 100.

Why truthful (no profitable deviation):

- Loser* raising b : still lose (pay 0) or win at price $> v_i$ (negative payoff). Lowering: still 0. $\Rightarrow$ no gain.
- Winner* raising b : still pays 2nd price (unchanged). Lowering: stays above 2nd $\rightarrow$ unchanged, or drops below $\rightarrow$ loses (0). $\Rightarrow$ no gain.

Equilibrium Concept & Revenue Equivalence

- 1st-price: **Bayesian-Nash** (incomplete info, values from distribution). 2nd-price: (**weakly dominant**) **Nash** for any realized values.
- Revenue Equivalence:** under risk-neutral bidders, iid private values from common distribution, symmetric eq., all 4 canonical formats give the seller the **same expected revenue**.
- Uses: 1st-price $\rightarrow$ privatization; 2nd-price $\rightarrow$ digital ad tech (Google, Facebook).

Coordination

Coordination: process where a group of agents choose a **single Pareto-optimal** NE. Many NE exist $\rightarrow$ real problem = **equilibrium selection**. **Pure coordination / team game**: all share one payoff: $u_i(a) = \dots = u_n(a) \equiv u(a)$. **Pareto dominance:** a dominates a' iff $\forall i: u_i(a) \geq u_i(a')$ and $\exists j: u_j(a) > u_j(a')$. **Pareto-optimal** = not dominated by any a' .

Social Conventions (Boutillier 1996) Recipe (common knowledge) constraining action choice so all pick the **same** NE; once established, no one gains by deviating.

- Fix a **unique ordering scheme of joint actions**: order **agents** + order **actions**.
- Each agent computes all equilibria.
- Each selects the **first** equilibrium in the ordering.

Must be common knowledge . If agents disagree on the ordering they may pick different NE (both cross $\rightarrow$ crash).

Ex (movie): order $1 \succ 2$, Thriller $\succ$ Comedy $\Rightarrow$ pick (Thriller,Thriller). Car right-of-way: right has priority.

With Communication Only need an **ordering of agents** (no full joint-action order). Sequential/synchronized:

- agent i waits until all $1, \dots, i-1$ broadcast their actions;
- picks a_i^* = component of a NE **consistent** with predecessors;
- broadcasts a_i^* to those not yet chosen.

Roles mask action set (deactivate actions) $\rightarrow$ shrink m , simplify selection. Assign greedily by **potential** (fit of agent to role) in fixed role order; one role/agent; $O(n^2)$.

Coordination Graphs (Guestrin 2002) reduce **number of agents**.

Joint actions = m^n (exp). **Roles cut m ; coordination graphs cut n .**

Global payoff = sum of local pairwise factors:

$$u(a) = \sum_{j=1}^k f_j(\cdot), \quad \text{e.g.} \quad u(a) = f_1(a_1, a_2) + f_2(a_1, a_3) + f_3(a_3, a_4)$$

Graph: nodes = agents, edges = local factors f_j . Solve many small subgames vs one huge one.

Variable Elimination (max over joint actions, exact)

- Pick agent i ; collect all factors f_j containing a_i .
- Compute its best response: $e_i(a_{\text{neigh}}) = \max_{a_i} \sum_j f_j$, recording $\arg \max a_i$.
- Replace those factors with new factor e_i over i 's neighbours; remove i .
- Repeat until one agent left $\rightarrow$ max gives optimal value; back-substitute $\arg \max$ to recover joint action a^* .

Roles	single reduced subgame, cuts m
Coord. graph	many subgames, cuts n , $u = \sum f_j$

Agent Architectures

Architecture: principles describing agent behavior: formal/abstract view + map of internals (control flow). Goal: guide agent design.

Abstract model Env states $E_o = \{e_0, e_1, \dots\}$; actions $\text{Ac} = \{a_0, \dots\}$. Run $r: e_0 \rightarrow e_1 \rightarrow e_2 \dots$:

- State transformer $\tau: R^{\text{Ac}} \rightarrow \text{pow}(E)$: history-dependent, non-deterministic; $\tau(r) = \emptyset \Rightarrow$ run ended.
- Env = $\langle E_o, \tau \rangle$. Agent $\text{Ag}: R^E \rightarrow \text{Ac}$ (maps run ending in state to action).

Signatures (trap): general $\text{Ag}: R^E \rightarrow \text{Ac}$ (history) vs purely reactive $\text{Ag}: E \rightarrow \text{Ac}$ (present only) vs $\tau: R^{\text{Ac}} \rightarrow \text{pow}(E)$. Deliberation fns use power sets 2^{Bel} , 2^{Des} , 2^{Int} .
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Deductive (symbolic) 1956-85 Explicit symbolic world model (predicate logic); decide by theorem proving $\text{DB} \vdash \text{Do}(a)$. 2 problems:

transduction (world->symbols), **representation/reasoning**. Bad: complexity, no time constraints, env can't change while deciding, hard for dynamic envs.

Deliberative / BDI (intentional stance)

Practical Reasoning = Deliberation (WHAT) + Means-Ends Reasoning (HOW)
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Beliefs (info), **Desires** (goals), **Intentions** (commitments). Intentions: persist, drive means-ends, constrain future deliberation (**filter of admissibility**), influence beliefs. Functions: $\text{brf}: 2^{\text{Bel}} \times \text{Per} \rightarrow 2^{\text{Bel}}$, options: $2^{\text{Bel}} \times 2^{\text{Int}} \rightarrow 2^{\text{Des}}$, filter: $2^{\text{Bel}} \times 2^{\text{Des}} \times 2^{\text{Int}} \rightarrow 2^{\text{Int}}$, plan: $2^{\text{Bel}} \times 2^{\text{Int}} \times 2^{\text{Ac}} \rightarrow \text{Plan}$. Loop: 1.observe+update beliefs 2.deliberate (options->filter)=intention 3.plan 4.execute 5.repeat. **Commitment (trap):** Blind=until achieved; Single-minded=until achieved OR impossible; Open-minded=while not achieved AND still desired. **Reactive (Brooks subsumption)** No symbolic reasoning; intelligence **distributed**, **emerges** from behavior+env interaction. Behavior modules = FSMs (action-selection); **lower layers inhibit higher** (lower = priority). Many fire at once. **Limits:** only local/short-term info; needs enough local info; **no learning**; hard to engineer complex behavior; **unpredictable with many layers**.

Hybrid (reactive+deliberative, reactive precedence) **Horizontal:** every layer touches input+output (mediator reconciles m^n ; conflict/coherence problem). **Vertical:** at most 1 layer per input/output (bottleneck, fault-intolerant); **one-pass** (up only) vs **two-pass** (up then down). Ex: **Stanley** DARPA 2005.

	Reactive	Deliberative	Hybrid
Reasoning	none/rules	symbolic/plan	both layered
Speed	fast	slow	real-time
Learning	no	—	layered abstraction
Weakness	short-term, simple	complexity,slow	layer interaction

Introduction to Agents

Agent: computer system capable of **autonomous action** in some environment to meet its **design objectives**, on behalf of its user/owner. **Autonomy** = freedom from external (human) control: figures out *what* to do. **MAS:** set of agents that **interact**, often for users with **different goals** $\Rightarrow$ need cooperation, coordination, negotiation.

- Loop:** sense (sensors/perceptors) $\rightarrow$ decide $\rightarrow$ act (effectors/actuators) $\rightarrow \dots$; closed-coupled, continual.
- Two problems:** agent design (micro: single-agent decision making) vs society design (macro: interaction).
- Autonomy levels:** operational (choose *actions*) vs motivational (choose *goals*). Bounded by: environment, resources, other agents, norms.

Three Intelligent-Agent Behaviours

- Reactivity:** maintains ongoing interaction with a **dynamic** environment, responds to changes *in time to be useful* (fixed env $\Rightarrow$ blind execution).
- Pro-activeness:** goal-directed; generates & attempts to achieve goals, takes initiative, not driven solely by events.
- Social ability:** interact with other agents via coordination/cooperation/negotiation + modeling others.

Open problem: balancing reactive (timely response) vs pro-active (long-term goals) — they can be at odds.

Social Abilities (define + example)

- Coordination** = managing interdependencies between activities. *Ex:* two agents share a non-sharable resource.
- Cooperation** = working as a team toward a **shared goal**. *Ex:* goal no single agent can achieve alone.
- Negotiation** = reach agreement on matters of **common interest** (possibly conflicting). *Ex:* roommates — he watches football tonight, you watch a movie tomorrow.

Trap: coordination = interdependencies; co-operation = shared goal; negotiation = agreement on conflicting interests. Don't conflate.
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Environment Properties (chess = “easy” side of all)

Di-chotomy	“Good” side	Contrast
Accessible/Inaccessible	complete, accurate, up-to-date state (= fully observable)	chess / real world
Deterministic/Non-det	action has single guaranteed effect	chess / football
Static/Dynamic	world unchanged while deliberating	chess / Counter-Strike
Discrete/Continuous	finite, countable actions/percepts	chess / landscape
Episodic/Non-episodic	independent intervals, no cross-influence	MRI scan / persistent game

Trap: accessible = fully observable (state); determinism = single action effect — distinct axes. Most complex envs are inaccessible + dynamic + non-deterministic.

- Other props:** adaptivity (learn), rationality (maximize utility fn), mobility (change location), curiosity, believability.

Markov Decision Processes

MDP: single-agent sequential decision making; **Markovian** (next s, r depend only on current s, a), **fully observable**, horizon may be infinite. Tuple (S, A, P, R, γ) : states, actions, transition $P(s'|s, a)$, reward $R(s, a)$, discount $\gamma \in [0, 1)$.

Value functions

$$V^\pi(s) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \mid s_0 = s, a_t = \pi(s_t) \right]$$

$Q^\pi(s, a)$: same but force first action a , then follow π . Optimal: $V^*(s) = \max_\pi V^\pi(s)$, Q^* analog; ≥ 1 optimal π^* always exists (may be many, all share V^*, Q^*). Greedy extract: $\pi^*(s) \in \arg \max_a Q^*(s, a)$.

Bellman optimality: $Q^*(s, a) = R(s, a) + \gamma \sum_{s'} P(s' s, a) \max_{a'} Q^*(s', a')$, $V^*(s) = \max_a Q^*(s, a)$.
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Solving (model known: need P, R)

- Value Iteration:** repeat $Q(s, a) := R(s, a) + \gamma \sum_{s'} P(s'|s, a) V(s')$ then $V(s) := \max_a Q(s, a)$ until stable; $\rightarrow Q^*$ for any init.
- Policy Iteration:** alternate *policy evaluation* (solve V^π) + *policy improvement* (greedy on Q^π) until π stable.

Markov / stochastic game Multiagent MDP: tuple $(S, \{A_i\}, P, \{R_i\}, \gamma)$; joint action $a = (a_1, \dots, a_n)$, $P(s'|s, a)$, each i has own R_i ; solution = equilibrium policies (e.g. Nash) not a single optimum.

Rational Agents

Rational agent: acts to **maximize (expected) utility**. Decisions are made **algorithmically** over the agent's preferences. **Preferences & utility** Relations on outcomes: strict $x \succ y$, indiff. $x \sim y$, pref-indiff. $x \succeq y$ ($= \succ$ or $\sim$). A preference is **rational** iff **complete** ($xRy \vee yRx$) and **transitive** ($xRy, yRz \Rightarrow xRz$); $\succeq$ is rational.

Utility thm: if $\succeq$ rational on X , exists $u: X \rightarrow \mathbb{R}$ with $u(x) \geq u(y) \Leftrightarrow x \succeq y$.

Decision rule No uncertainty: $Q(a) = u(O(a))$. Under uncertainty: maximize **expected utility**

$$a^* = \arg \max_{a \in A} \sum_{o \in O} u(o) P(o \mid a)$$

PEAS: task-environment spec: **P***erformance measure, **E**nvironment, **A***cttuators, **S***en-

sors. **Decision tree:** *decision* nodes (diamond, agent chooses) + *chance* nodes (circle, prob.); eval each branch by exp. utility, pick max.

Hybrid MAS (rarely tested)

Hybrid MAS: collective of **human** + **AI** agents sharing an env; both sense/act, have goals. **Layering:** reactive+deliberative layers: **horizontal**=all layers see input (needs mediator, e.g. TouringMachines), fault-tolerant/parallel; **vertical**=pipeline (e.g. InterRRaP).

- Delegation** = control $\rightarrow$ influence; user gives **intent**, agent fills details. Trust dims: Competence, Predictability, Benevolence, Integrity. Autonomy: in-loop / on-loop / out-of-loop.
- Contract-Net:** manager broadcasts call-for-proposals $\rightarrow$ agents bid/decline $\rightarrow$ award lowest-cost bid.

Voting (not seen on past exams)

Social choice fn: maps all agents' (truthful) preferences $\rightarrow$ one enforced outcome. **Schemes** **Plurality:** most first-place votes wins. **Plurality+elim (IRV):** drop fewest-votes, transfer to next, repeat. **Borda:** top gets $n-1$, next $n-2, \dots$, last 0 (n =#cands); sum over voters, max wins. **Pairwise elim:** fixed agenda; majority runoffs eliminate loser. **Condorcet winner:** beats every other candidate in pairwise majority. Desirable but **need not exist:** cycle $A \succ B, B \succ C, C \succ A$ (Condorcet paradox).

Profile 499: $A \succ B \succ C, 3 \cdot B \succ C \succ A, 498 \cdot C \succ B \succ A \rightarrow$ Plurality= A (499), IRV= C (B out, $3 \cdot C \succ 501$), Condorcet= B (beats A 501-499, C 502-498). 3 methods, 3 winners.
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Defects IIA violated: dropping a losing candidate flips winner (Borda/plurality). **Agenda:** in pairwise elim the order alone picks the winner (last-compared has edge). **Pareto:** pairwise elim can pick D even though all prefer $B \succ D$.

Arrow: with ≥ 3 candidates, Pareto + IIA $\Rightarrow$ dictatorship. No fair non-dictatorial ranked system has both. (Pareto: all prefer $A \succ B \Rightarrow$ B not winner; IIA: pairwise pref independent of other alternatives.)

Gibbard-Satterthwaite ≥ 3 outcomes: any non-dictatorial, onto social choice fn is **manipulable** (some voter gains by lying). Truthful + non-dictatorial impossible. **Seats:** D'Hondt quotient $\frac{V_i}{s_i+1}$, s =seats won so far (not total!); largest quotient wins next seat. Sainte-Laguë: divisors 1,3,5,... (fairer to small parties); Hare-LR: largest remainder, most proportional. Gerrymandering: boundaries can let minority win.